Kora Network Architecture Addendum

v2.1 - Companion to Kora Rails Build Spec v2.0.0
Restores Kora's network definition without replacing the lean runtime architecture.

Canonical product definition. Kora is African financial interoperability infrastructure connecting financial institutions, fintechs, banks, mobile-money networks, Stellar anchors and liquidity providers through Stellar, including on-ramp, off-ramp, cross-border settlement and Stellar asset purchasing.

This document is additive. Unless explicitly amended below, the ledger invariants, Payment Intent model, provider connector contract, idempotency rules, outbox pattern, reconciliation loop, security baseline, testing strategy and phased discipline in Build Spec v2.0.0 remain canonical.

1. The correction this addendum makes

Build Spec v2.0.0 correctly makes Kora's internal double-entry ledger the canonical record of money and correctly avoids premature microservices. The product definition, however, is narrower than the intended company: it reads primarily as a PSP/payment orchestrator with Stellar introduced as one later corridor. Kora's target is broader - a reusable interoperability network across African financial rails.

Keep: ledger-first accounting; four deployable systems; rail-type connectors; Payment Intents; exactly-once payout instructions; compliance on the write path; reconciliation; no Kafka/Kubernetes/Soroban unless an actual scale or product requirement justifies them.

Restore: Kora Network as the product; Stellar as the primary shared external settlement fabric for cross-border and digital-asset corridors; a graph-shaped routing domain; first-class Stellar asset purchasing; institutional participants and liquidity providers as network members.

2. Product model: Kora Network

Kora accepts instructions to move value between financial endpoints. Endpoints may be bank accounts, mobile-money wallets, merchant ledger balances, Stellar accounts or supported Stellar assets. Kora normalizes those endpoints, determines an executable route, records obligations on its internal ledger, coordinates providers, settles the appropriate legs, reconciles external evidence and exposes one consistent API to merchants and institutions.

The network has four first-class capabilities: on-ramp (local fiat to supported Stellar assets), off-ramp (supported Stellar assets to local fiat), cross-border settlement (local rail in one market to local rail in another, commonly using Stellar in the middle), and Stellar asset access (purchase/redemption of allowlisted Stellar-issued assets using supported local rails).

3. Runtime architecture stays lean

Do not create a Routing Service, Liquidity Service, Treasury Service, Stellar Service, Asset Service, FX Service and Notification Service merely because they are distinct domains. They remain modules in

the existing v2 runtime until a named SLA, ownership boundary or throughput constraint forces a split.

```
CLIENTS
Business Dashboard | Hosted Checkout | Docs | Merchant/Institution Apps
                          |
                    API Gateway
                          |
        +-----------------+-----------------+
        |                                   |
  payments-api                        platform-api
  - intents                           - merchants
  - network graph                     - API keys
  - route planning                    - webhooks
  - quotes / FX                       - access policy
  - asset catalog
  - treasury views
        |
  payments-worker
  - outbox / retries
  - route execution
  - provider instructions
  - Stellar settlement orchestration
  - notifications
        |
  +-----+----------------+-------------------+
  |                      |                   |
ledger module       compliance module     connectors
                                         banks/
                                         mobile-money/
                                         stellar-anchor/
                                         fx-liquidity/
                                         stellar-network/
```

Deployable count remains effectively the v2 model: API, worker and web applications, with ledger and compliance beginning in-process and split only under the existing exit criteria. The additional Kora Network capabilities are domain modules and connector capabilities, not an excuse for service sprawl.

4. Canonical ledger vs. Stellar settlement

Kora ledger is canonical for customer, merchant and operational balances. Provider callbacks, bank statements, mobile-money receipts and Stellar transactions are external evidence projected onto that ledger through controlled journal postings.

Stellar is Kora's primary shared external settlement fabric for cross-border and digital-asset corridors. It is not the system of record, and not every local payment must touch Stellar. A domestic NGN collection can remain a local-rail transaction. A NGN-to-KES corridor, an NGN-to-USDC purchase or USDC-to-GHS payout can use Stellar as the common settlement leg.

This distinction preserves operational correctness while making Stellar structurally important to Kora's network thesis. The system must degrade safely if Stellar or an anchor is unavailable: local operations that do not require that settlement leg may continue; affected intents enter a pending/blocked settlement state and resume through the worker after health and reconciliation checks.

5. Expand the domain model, not the service count

Domain	Add to v2	Lives in v1/v2 runtime
Network Participant	bank, fintech, MoMo operator, anchor, LP, FX provider, processor	platform/providers schema + connector registry
Asset	fiat, Stellar classic asset, native XLM; code + issuer + network + scale + allowlist status	payments-api asset registry
Rail Endpoint	bank account, MoMo wallet, Stellar account, internal Kora account	payments schema
Network Edge	provider capability joining source endpoint/asset to destination endpoint/asset	routing module in payments-api
Route Plan	ordered route legs, quote snapshot, constraints and fallback policy	payments-api + worker
Liquidity Snapshot	available/reserved capacity by provider, asset and corridor	worker module + Postgres/Redis cache
Asset Purchase	fiat-funded purchase/redemption of an allowlisted Stellar asset	Payment Intent subtype/capability, not a new service

6. Kora routing graph

Keep v2's simple route-selection implementation initially, but make the domain graph-shaped now. Currencies, assets and rail endpoints are nodes. Provider capabilities are directed edges. This prevents the first provider implementation from hard-coding Kora into a pairwise country model.

Each edge should be able to describe: provider; source asset/rail; destination asset/rail; capability (collect, payout, FX, on-ramp, off-ramp, settlement); fees; effective rate; min/max; estimated latency; health; success history; liquidity/capacity; compliance constraints; operating window; quote expiry; and whether the edge is currently executable.

```
Example route plan

NGN bank
-> local collect connector
-> Kora ledger obligation
-> NGN/USDC liquidity edge
-> Stellar USDC settlement
-> destination anchor / LP
-> KES payout connector
-> M-Pesa beneficiary

The merchant sees one Payment Intent and one Kora status lifecycle.
Provider identities do not need to leak into the public API.
```

7. Routing policy

Do not build a sophisticated graph optimizer in Phase 0. Start deterministic: filter invalid edges, check compliance and liquidity, then choose among executable candidates using a small policy. Persist the chosen RoutePlan before external execution so every leg can be reconciled and recovered.

Route scoring may later include total delivered amount, fee, FX spread, provider health, success rate, latency and liquidity. Strategies such as cheapest, fastest, most reliable and balanced are product features only after Kora has enough real execution data to make those scores meaningful.

8. Liquidity and treasury inside the existing topology

The existing v2 liquidity check remains correct for the first rail. Extend it into a network-aware liquidity module rather than a new deployable. Track external float and Kora reservations by provider, asset and corridor. A quote or approved payout may reserve capacity for a short TTL. The worker releases or consumes the reservation as execution progresses.

Treasury views should answer: current Kora ledger exposure by currency; external float by provider; available corridor capacity; pending Stellar settlement; stale/unreconciled balances; and which provider requires manual rebalancing. Automatic netting, hedging and autonomous rebalancing are explicitly later.

9. Stellar integration model

Retain the existing connectors/stellar-anchor approach for anchor-mediated deposit, withdrawal and cross-border flows. Add connectors/stellar-network only as a rail connector/library for direct Stellar payment submission, transaction observation, asset/trustline operations where required and settlement confirmation. It is not a standalone microservice.

Kora should prefer established anchor/custody infrastructure and Stellar payment primitives over custom Soroban contracts. Contracts are introduced only where Kora owns a genuinely programmable policy that cannot be expressed safely through existing rails.

Settlement matching rule: a Stellar transaction is evidence, not a ledger mutation by itself. The worker matches network/anchor evidence to the expected RouteLeg, amount, asset issuer, destination and reference, then posts the corresponding journal through the existing ledger interface.

10. First-class Stellar asset purchasing

Kora must support more than using USDC internally. A merchant, fintech or Kora first-party client may request purchase or redemption of an allowlisted Stellar asset using a supported local rail. This remains inside the Payment Intent model rather than creating a separate exchange product.

Recommended additions to the public API:

```
GET /v1/assets
GET /v1/assets/:id
GET /v1/assets/:id/availability?country=NG
POST /v1/asset-quotes
POST /v1/intents type=asset_purchase
POST /v1/intents type=asset_redemption
GET /v1/intents/:id
```

An asset identity must include network and issuer, not merely a ticker. The registry should carry verification/allowlist status, decimals, deposit/withdraw availability, supported source countries and the connector paths that can execute the purchase.

11. Cross-border settlement becomes a first-class Intent outcome

Do not require merchants to manually create an on-ramp, Stellar transfer and off-ramp. The public instruction should express source and destination. Kora constructs the internal RoutePlan.

```
POST /v1/intents
{
  "type": "transfer",
  "source": {
    "amount": {"currency": "NGN", "minor": "50000000"},
    "instrument": "bank_transfer"
```

```
  },
  "destination": {
    "currency": "KES",
    "instrument": "mobile_money",
    "beneficiary_id": "bnf_..."
  }
}

Kora may execute:
NGN collect -> FX/on-ramp -> Stellar settlement -> off-ramp -> KES M-Pesa

The API contract remains one intent.
```

12. Connector contract extensions

Keep the existing Connector interface. Extend capabilities rather than creating provider-specific services. Optional methods can expose quotes, liquidity/capacity and asset support.

```
interface Connector {
  id: string;
  rails: Rail[];
  capabilities: Capability[];

  getCapabilities?(): Promise;
  getQuote?(req: QuoteRequest): Promise;
  getLiquidity?(req: LiquidityRequest): Promise;

  resolveInstrument?(req: ResolveRequest): Promise;
  collect(req: CollectRequest): Promise;
  payout(req: PayoutRequest): Promise;
  refund?(req: RefundRequest): Promise;

  parseWebhook(...): Promise;
  fetchStatus(providerRef: string): Promise;
  fetchStatement?(...): Promise;
}
```

13. Institutional participation

Kora's merchant model remains valid, but the platform domain should recognize that a Kora customer can also be a fintech or financial institution consuming the network. A bank, wallet, remittance company or fintech may use Kora APIs for collection, payout, asset access or cross-border settlement without adopting Kora's consumer brand.

Do not build separate applications for every participant type. Extend merchant/business metadata with organization type, licenses/permissions, enabled corridors, settlement configuration, fee schedule and API scopes. Add specialized institutional workflows only when contracts require them.

14. Public surfaces and SDKs

The v2 launch surface remains: Public API, Business dashboard, Docs and hosted checkout. TypeScript remains the first SDK. Python follows when institutional integration demand exists. The consumer mobile app, React Native SDK, MCP and AI skills remain deferred. The important correction is that the API model must already represent Kora Network concepts even if the first UI exposes only a subset.

The first-party Business dashboard and any future Kora consumer/mobile product must consume the same public-domain payment primitives used by external integrators. Avoid a privileged parallel payment stack.

15. MCP and agent interfaces remain later

Do not scaffold MCP merely for feature completeness. Once reconciliation, authorization and payout safety are proven, Kora MCP can expose read operations (balances, assets, quotes, routes, transaction status) and tightly authorized write operations. Money-moving agent permissions must be constrained by scopes, amount limits, approved beneficiaries/corridors and human approval thresholds.

16. Revised phased delivery

Phase	Deliverable	Exit condition
0 - Financial core	Money library, ledger, invariants, fake connector, schemas	Invariant/property tests green
1 - Payment engine	Payment Intent, worker, outbox, webhook, hosted fake collect	End-to-end fake payment survives retry/crash
2 - Nigeria rails	Real NGN collect + payout, name enquiry, compliance/recon	Sandbox/live pilot, zero duplicate instructions
3 - Production core	Ops, reconciliation, KYB, TS SDK, docs, security hardening	First regulated live merchant
4 - Stellar + assets	Anchor/network connector, USDC settlement, asset registry, asset purchase/redemption	Closed NGN <-> Stellar asset loop
5 - First cross-border	Second African rail, cross-border transfer Intent, route plan	Local currency A -> Stellar -> local currency B completes/reconciles
6 - Kora Router	Multiple executable providers/edges, health/liquidity-aware selection, fallback	At least one corridor has real route choice and safe recovery
7 - Network expansion	Additional countries, LPs, institutional integrations, Python SDK	Corridor reliability metrics meet targets
8+ - Optional surfaces	Consumer/mobile, MCP/skills, advanced treasury/netting as demand proves them	Named product/scale requirement

17. Revised repository delta

Do not replace the v2 monorepo. Make only these conceptual additions:

```
packages/
  network/          # nodes, edges, RoutePlan, route constraints
  assets/           # Stellar/fiat asset registry and allowlist types
  # money/, provider-sdk/, ledger etc. remain as v2

connectors/
  banks/
  mobile-money/
  stellar-anchor/
  stellar-network/  # direct Stellar rail helper; not a service
  fx-liquidity/     # quote/liquidity connectors grouped by capability

specs/
  network/model.md
  assets/registry.md
  routing/policy.md

NO new routing/, liquidity/, treasury/, stellar/ deployable services.
```

18. Revised success metrics

Keep all v2 financial safety metrics. Add network metrics only when the relevant phase exists:

• Closed-loop local-to-Stellar and Stellar-to-local settlement reconciliation: 100% accounted for.

• Cross-border terminal accuracy vs. provider/chain evidence: >99.9% matched within the defined reconciliation window.

• Duplicate provider payout instruction rate: 0.

• Asset purchase issuer mismatch / wrong-asset execution rate: 0.

• Route plan recovery: an ambiguous provider timeout must never trigger an unsafe blind duplicate.

• Provider/rail health and liquidity visibility exists before automatic failover is enabled.

• Do not open the next corridor while ledger invariants, reconciliation or duplicate-payment metrics are degraded.

19. Canonical decision log amendments

Decision	Canonical choice
System of record	Internal double-entry ledger. Never infer customer balances from providers or Stellar.
Stellar	Primary shared external settlement fabric for cross-border and digital-asset corridors; not the application ledger.
Kora product	African financial interoperability network, not merely a domestic PSP or ramp aggregator.
Routing	Graph-shaped domain model from the start; simple deterministic policy first; no separate service until an exit criterion exists.
Assets	Stellar asset purchase/redemption is first-class. Every asset is identified by network + issuer + code and allowlisted.
Services	Preserve the lean v2 topology. New capabilities are modules/connectors before they are services.
Soroban	Optional. Use only for a concrete programmable policy not safely handled by existing Stellar/anchor primitives.
MCP/mobile	Deferred until payment/reconciliation reliability and product demand justify them.

20. Definition of 'Kora exists'

When the v2 first ten tickets are green, Kora's financial core exists. Kora Network exists when that core has proven at least one local rail, one Stellar settlement/asset path and one cross-border path into a second African rail, all through the same ledger, Payment Intent, connector and reconciliation architecture.

North star: an integrator should be able to specify a source of value and a destination of value - for example, an NGN bank rail and a KES mobile-money beneficiary - and Kora should determine the valid providers, compliance requirements, liquidity, Stellar settlement path, execution sequence and recovery behavior while exposing one coherent transaction lifecycle.

Implementation instruction: read this addendum together with Kora Rails Build Spec v2.0.0. Where the documents differ on product scope or Stellar's role, this addendum governs. Where they differ on ledger safety, idempotency, reconciliation, connector discipline, runtime topology or premature infrastructure, v2.0.0 remains authoritative unless explicitly amended above.